

Q1.

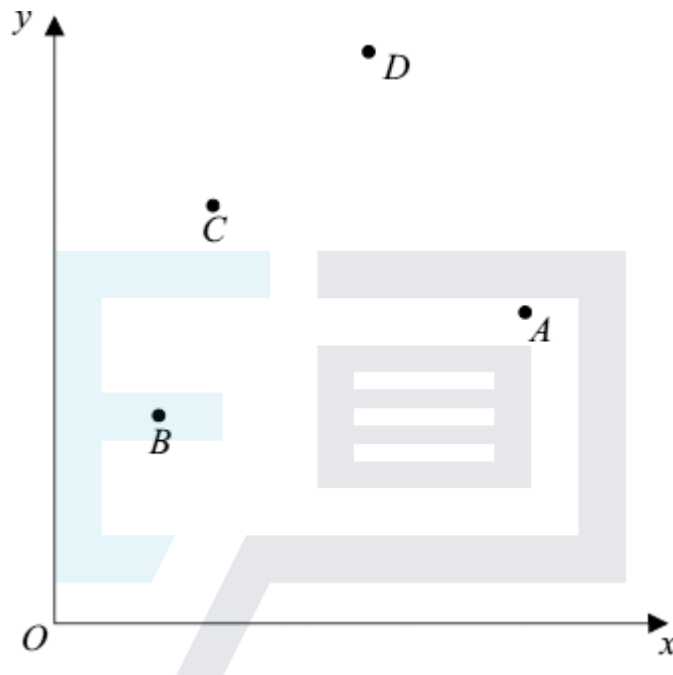
The diagram shows four particles, A , B , C and D , which are fixed in a horizontal plane which contains the x - and y -axes, as shown.

Particle A has mass 2 kg and is attached at the point $(9, 6)$.

Particle B has mass 3 kg and is attached at the point $(2, 4)$.

Particle C has mass 8 kg and is attached at the point $(3, 8)$.

Particle D has mass 7 kg and is attached at the point $(6, 11)$.



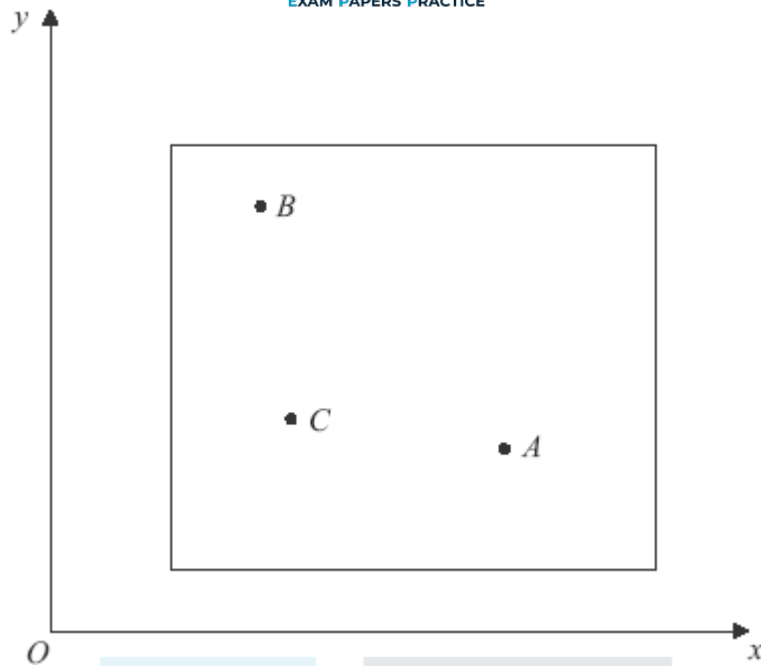
Find the coordinates of the centre of mass of the four particles.

(Total 5 marks)

Q2.

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A piece of modern art is modelled as a uniform lamina and three particles. The diagram shows the lamina, the three particles A , B and C , and the x - and y -axes.



The lamina, which is fixed in the x - y plane, has mass 10 kg and its centre of mass is at the point (12, 9).

The three particles are attached to the lamina.

Particle A has mass 3 kg and is at the point (15, 6).

Particle B has mass 1 kg and is at the point (7, 14).

Particle C has mass 6 kg and is at the point (8, 7).

Find the coordinates of the centre of mass of the piece of modern art.

(Total 6 marks)

Q3.

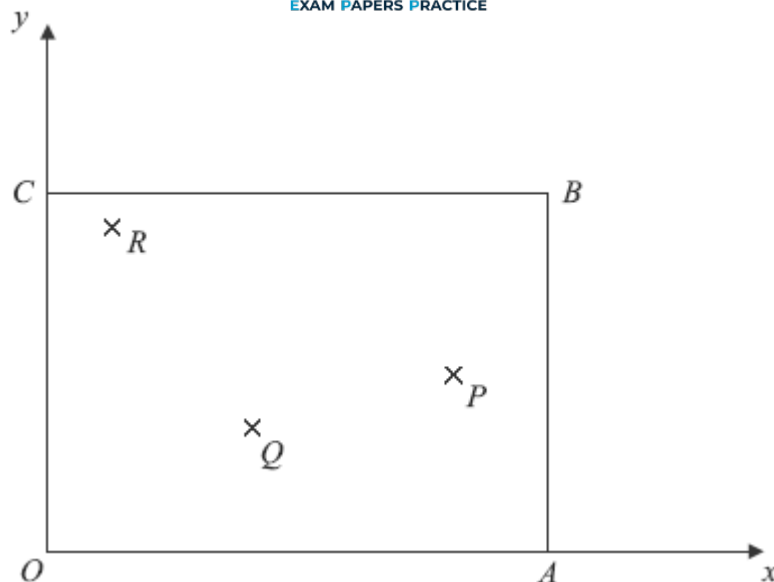
Three particles are attached to a light rectangular lamina $OABC$, which is fixed in a horizontal plane.

Take OA and OC as the x - and y -axes, as shown.

Particle P has mass 1 kg and is attached at the point (25, 10).

Particle Q has mass 4 kg and is attached at the point (12, 7).

Particle R has mass 5 kg and is attached at the point (4, 18).



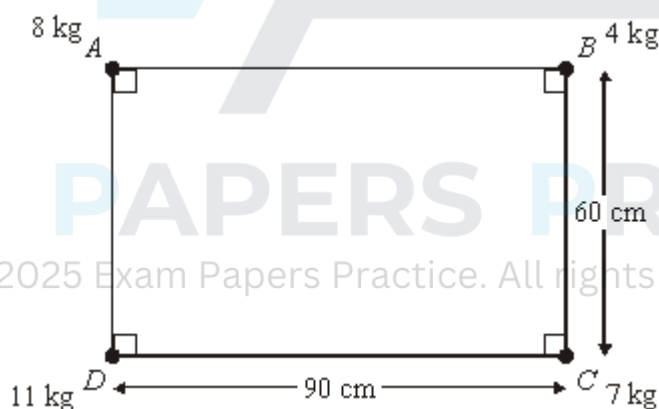
Find the coordinates of the centre of mass of the three particles.

(Total 4 marks)

Q4.

Particles of masses 8 kg, 4 kg, 7 kg and 11 kg are attached to the vertices A , B , C and D respectively of a light, rigid, rectangular framework $ABCD$.

In the framework, $CD = 90$ cm and $BC = 60$ cm, as shown in the diagram.



(a) Show that the centre of mass of the system of particles is 33 cm from AD .

(3)

(b) Find the distance of the centre of mass of the system of particles from AB .

(3)

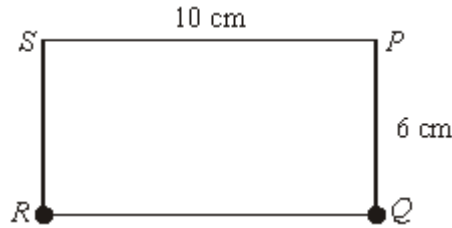
(c) The framework is freely suspended from the corner A and hangs in equilibrium. Find the angle, in degrees, between the downward vertical and the side AB .

(3)

(Total 9 marks)

Q5.

A uniform rectangular plate, $PQRS$, has mass 1 kg. Particles of mass m kilograms are attached to the plate at Q and R . The plate is shown in the diagram below.



The dimensions of the plate are $PQ = SR = 6$ cm and $PS = QR = 10$ cm.

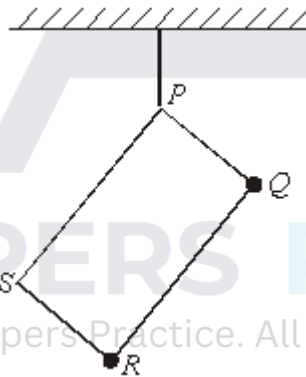
- (a) State the distance of the centre of mass of the system from PQ . (1)

- (b) Show that the distance, in centimetres, of the centre of mass of the system from PS is

$$\frac{12m + 3}{2m + 1}.$$

(3)

- (c) The plate is freely suspended from P , as shown in the diagram below.



When the plate hangs in equilibrium, PS makes an angle of 45° with the downward vertical. Find the value of m .

(5)

(Total 9 marks)

Q6.

Two particles, of masses 2 kilograms and m kilograms, are placed at the points $(1, 8)$ and $(7, 11)$, respectively, in the x - y plane. The unit of distance is the metre.

The centre of mass of this system of particles lies on the line $x = 5$.

- (a) Show that $m = 4$. (3)

- (b) Determine the y -coordinate of the centre of mass of this system.

(3)
(Total 6 marks)

Q7.

A uniform rod AB has length 4 m. Three particles are attached to the rod as listed below.

Mass of particle	Distance from A
3 kg	1 m
12 kg	2.5 m
15 kg	3 m

The mass of the rod is 10 kg.

- (a) Show that the centre of mass of the rod and the particles is 2.45 metres from A .

(4)

- (b) Ropes are attached to each end of the rod. Find the tension in each rope if the rod is supported in equilibrium by the ropes, with the rod horizontal and the ropes vertical.

(5)

(Total 9 marks)

Q8.

The diagram shows a light square lamina. Particles with masses as shown in the diagram are fixed at the corners and centre of the lamina. The sides of the square are of length 1.2 metres.



- (a) Find the distance of the centre of mass of the system of particles from the side AB .

(3)

- (b) Explain why the distance of the centre of mass from the side AD is 0.6 metres.

(1)

- (c) The lamina is suspended in equilibrium from the corner A . Find the angle between the side AB and the vertical.

(4)



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